

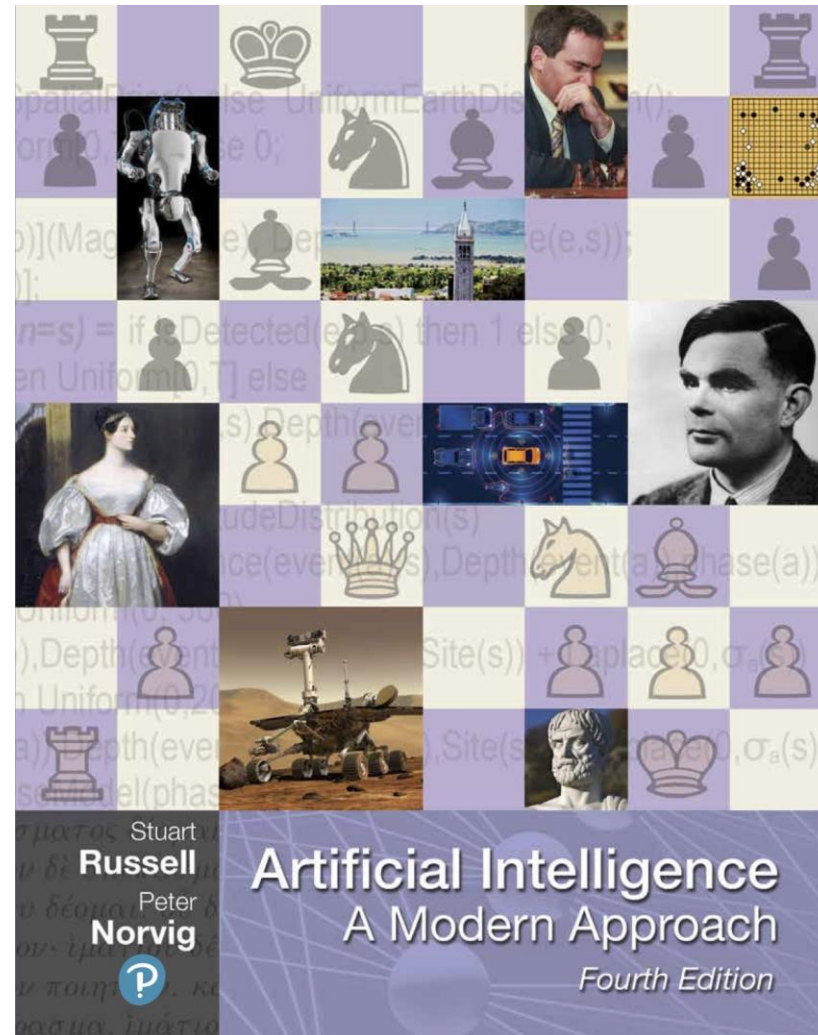
Artificial Intelligence

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- Textbook: Artificial Intelligence A Modern Approach
By Stuart Russell and Peter Norvig



Lecture 1

- What is AI?
- Intelligent agents
- Agent program
- Properties of environments

What is Artificial Intelligence (AI) ?

- AI encompasses various subfields, ranging from the general to the specific, such as playing chess, writing poetry, driving a car, and diagnosing diseases.
- The definition of AI can be organized into four categories.

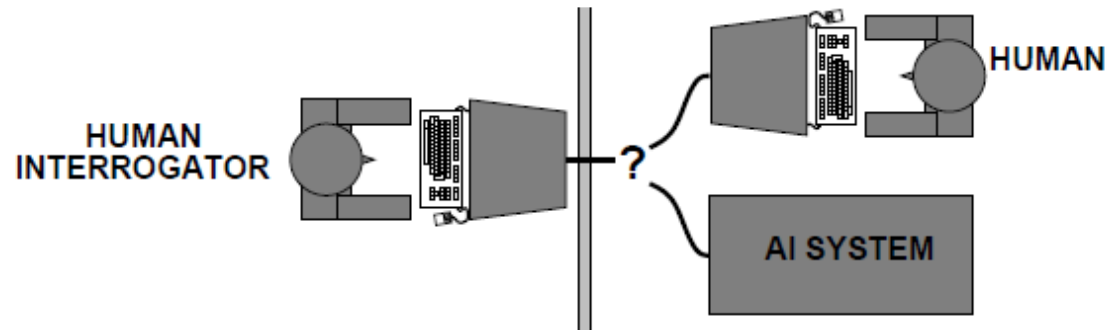
Thinking Humanly	Thinking Rationally
Acting Humanly * Need the Turing test	Acting Rationally

Acting Humanly: The Turing Test approach

- The Turing Test, proposed by Alan Turing (1950), was designed to provide a satisfactory operational definition of intelligence.



- A computer passes the test if a human interrogator cannot tell whether the written responses come from a person or a computer.

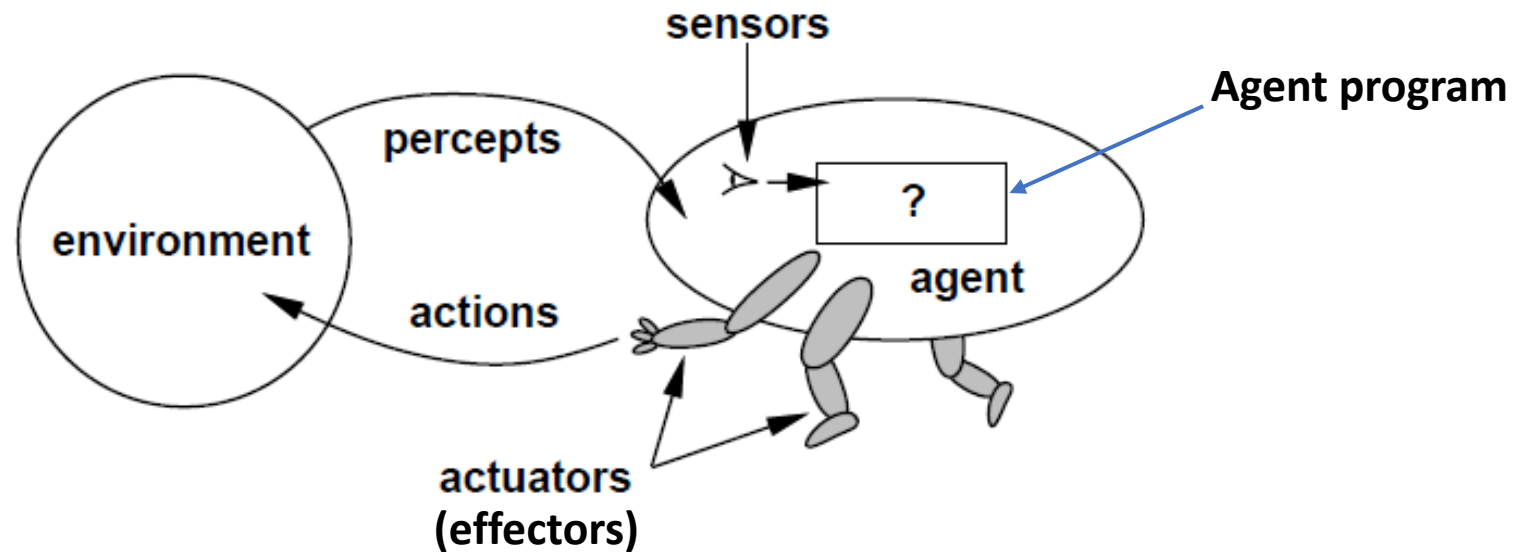


To take the Turing Test, the computer needs the following capabilities:

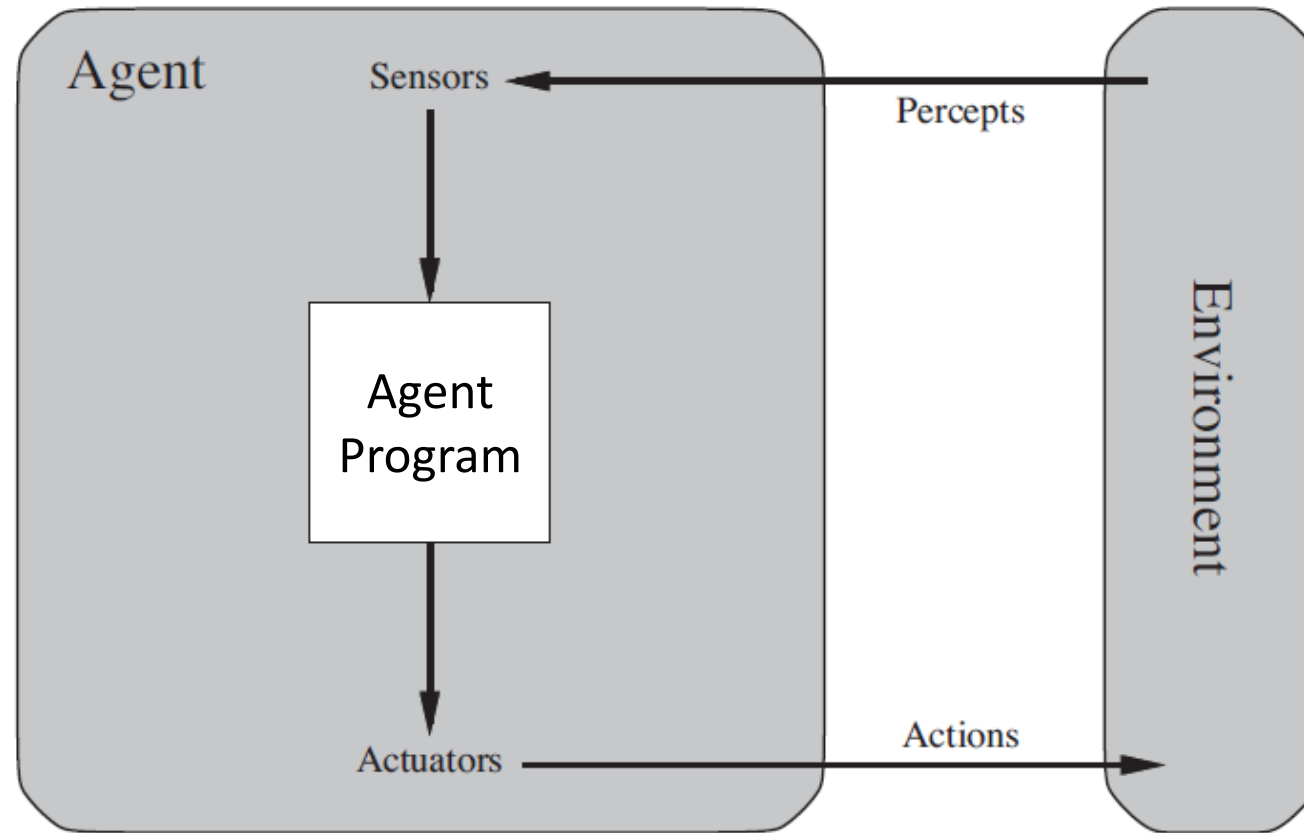
- Natural language processing
- Knowledge representation (transform fact to sentence)
Ex. Today is hot $\Rightarrow$ IS(Today, Hot)
- Automated reasoning
Ex. Bigger(A, B) and Bigger(B,C) $\Rightarrow$ Bigger(A,C)
- Machine learning

Intelligence Agents

Agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through effectors.



Overview of Agents



- We would need our agent to be rational.
- A rational agent always does the right thing (most successful).
- Rationality depends on four things
 1. Performance measure.
 2. Everything that the agent has perceived so far (**percept sequence**).
 3. What the agent knows about the environment.
 4. The actions that the agent can perform.

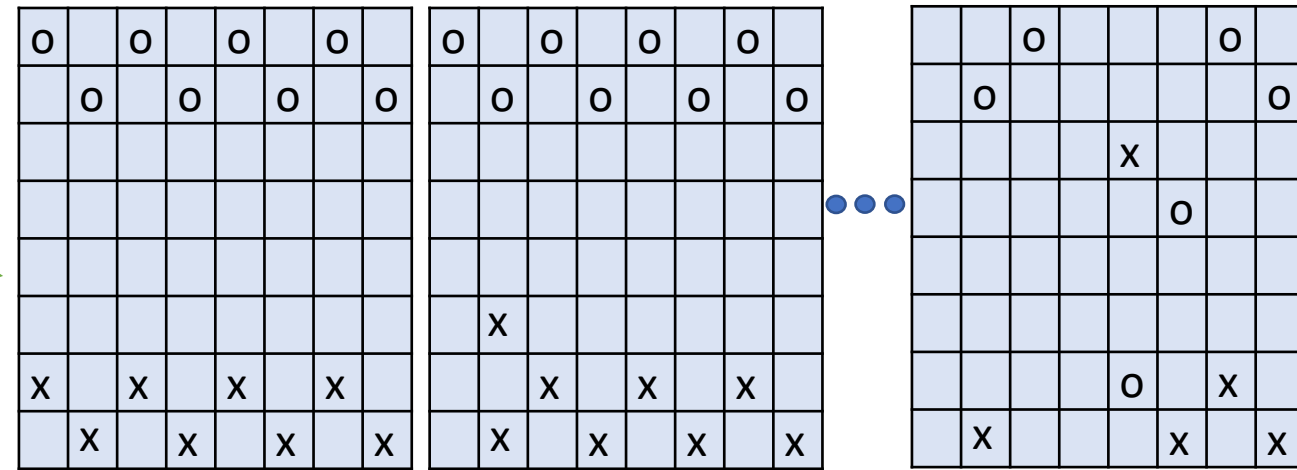
Agent Program

0) Table lookup (won't be counted as agent program)

Keeping in memory its entire percept sequence, and using it to index into table, which contains the appropriate action for all possible percept sequences.

Percept sequence	Action
P3, P1, P4, P2, P1, P4, P2, ..., P _i	A2
P1, P4, P6, P2, P3, P5, P3, ..., P _j	A4
P3, P1, P7, P10, P3, P4, P2, ..., P _k	A1
P2, P6, P4, P2, P1, P4, P9, ..., P _n	A3
...	...

Example: Checker



P1

P4

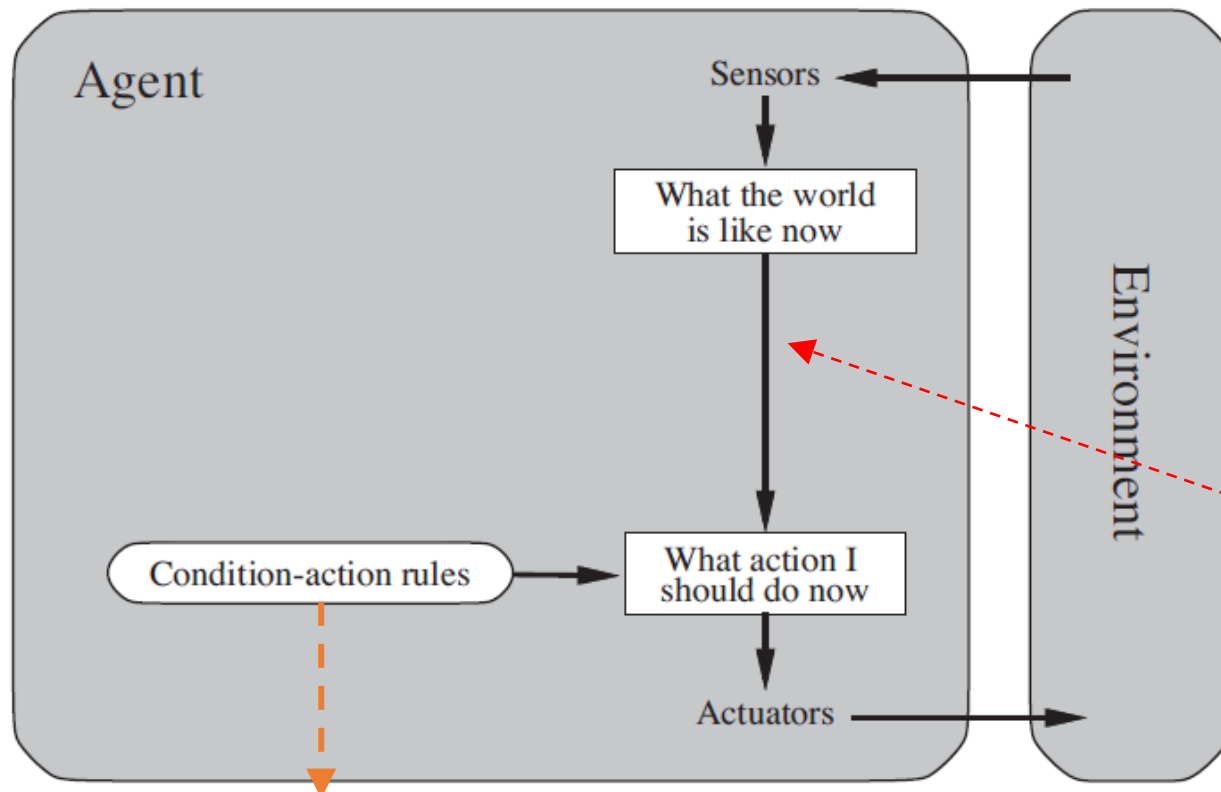
P_j

Disadvantages

1. Space (Ex: Chess need 35^{100} entries $\approx 2.55E+154$)
2. Time (Construction & Execution)
3. No autonomy (If the percept sequence doesn't perfectly match, it take no action)
4. Take forever for agent to learn the table entries

Agent Program

1. Simple Reflex Agents



IF car-in-front-is-braking **THEN** Initiate-braking
State (defined by percept)

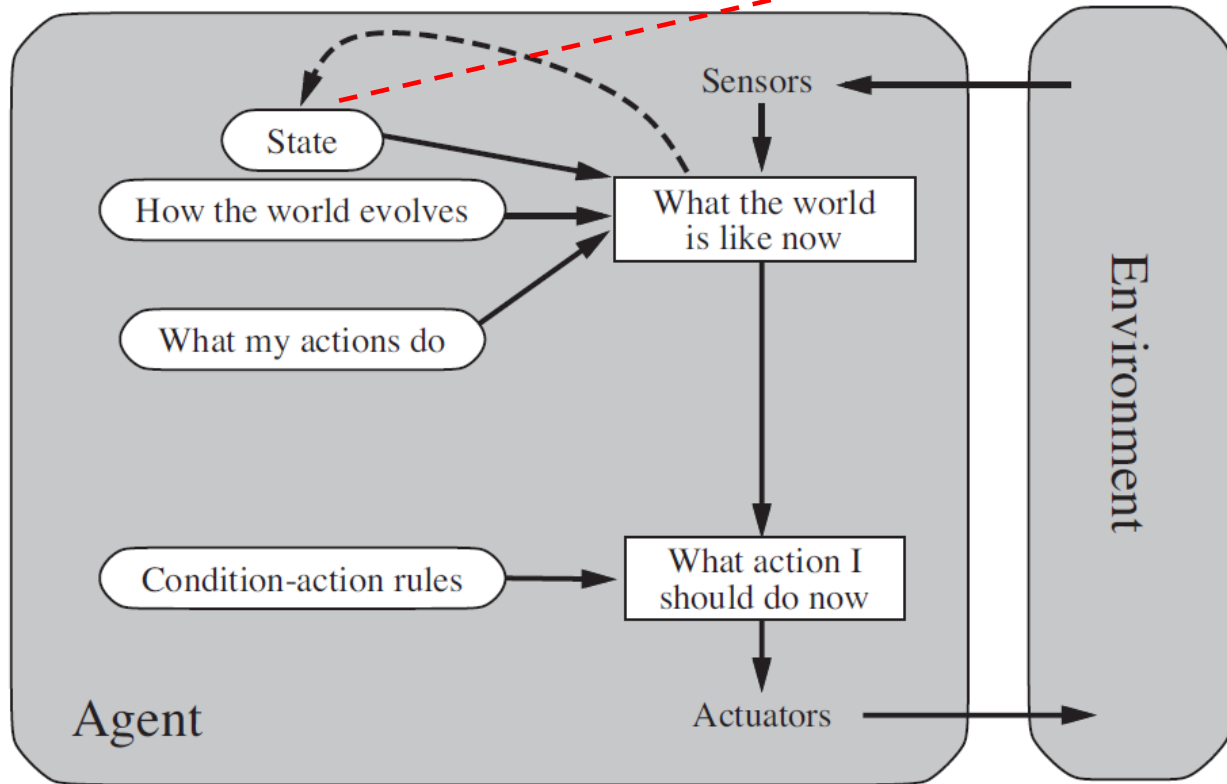
A simple reflex agent acts according to a rule whose condition matches the current percept.

function SIMPLE-REFLEX-AGENT(*percept*) **returns** an action
persistent: *rules*, a set of condition–action rules

state $\leftarrow$ INTERPRET-INPUT(*percept*)
rule $\leftarrow$ RULE-MATCH(*state*, *rules*)
action $\leftarrow$ *rule*.ACTION
return *action*

Agent Program

2. Model-based Reflex Agents



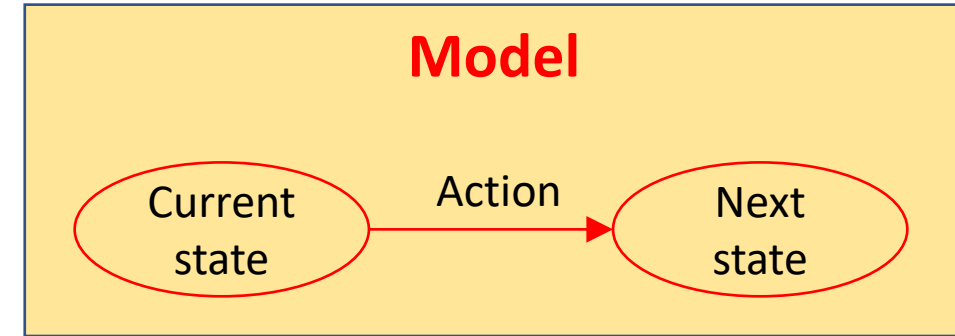
- Some problems need to handle partial observability from the past (**internal state**) which cannot see now.
- For the braking problem, we need the previous frame from the camera, allowing the agent to detect when two red breaking lights of the car go on/off simultaneously.

Agent Program

2. Model-based Reflex Agents (Cont.)

function MODEL-BASED-REFLEX-AGENT(*percept*) **returns** an action
 persistent: *state*, the agent's current conception of the world state
 model, a description of how the next state depends on current state and action
 rules, a set of condition–action rules
 action, the most recent action, initially none

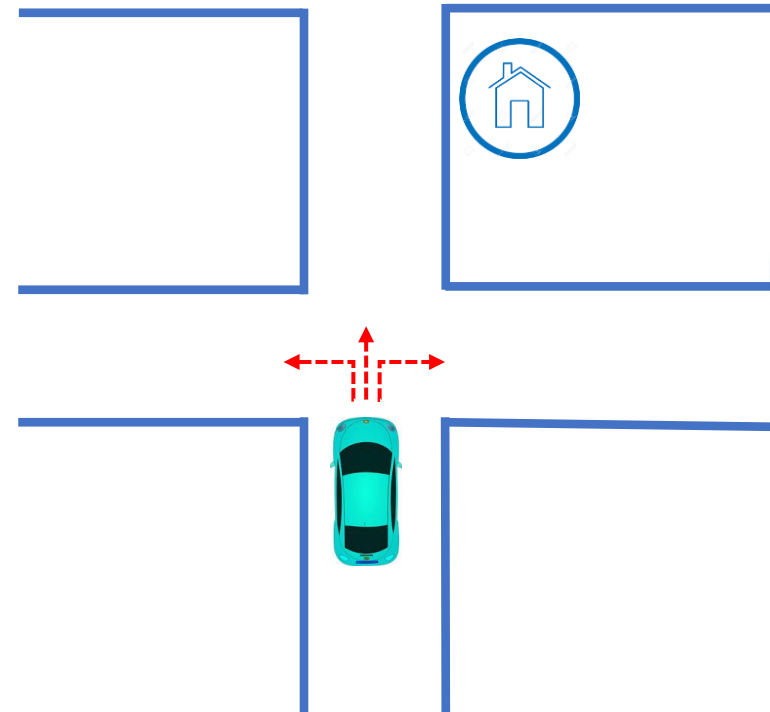
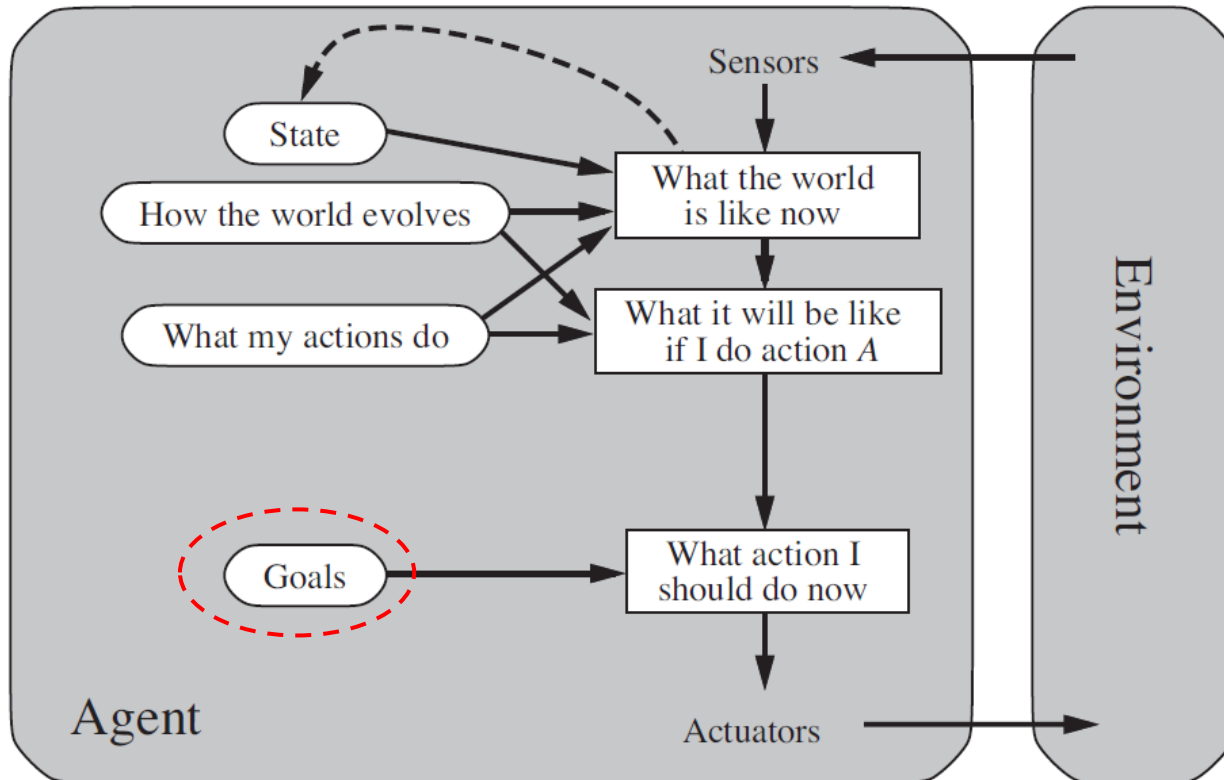
state ← UPDATE-STATE(*state*, *action*, *percept*, *model*)
rule ← RULE-MATCH(*state*, *rules*)
action ← *rule*.ACTION
return *action*



Agent Program

3. Goal-based Agents

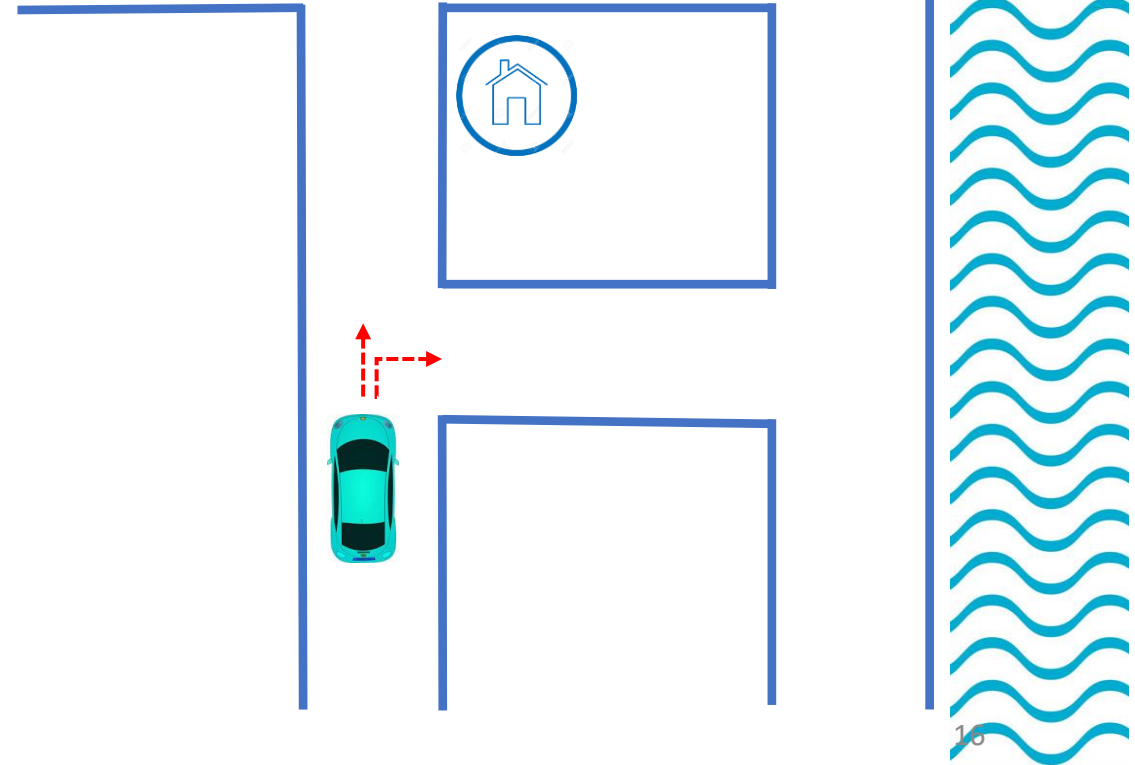
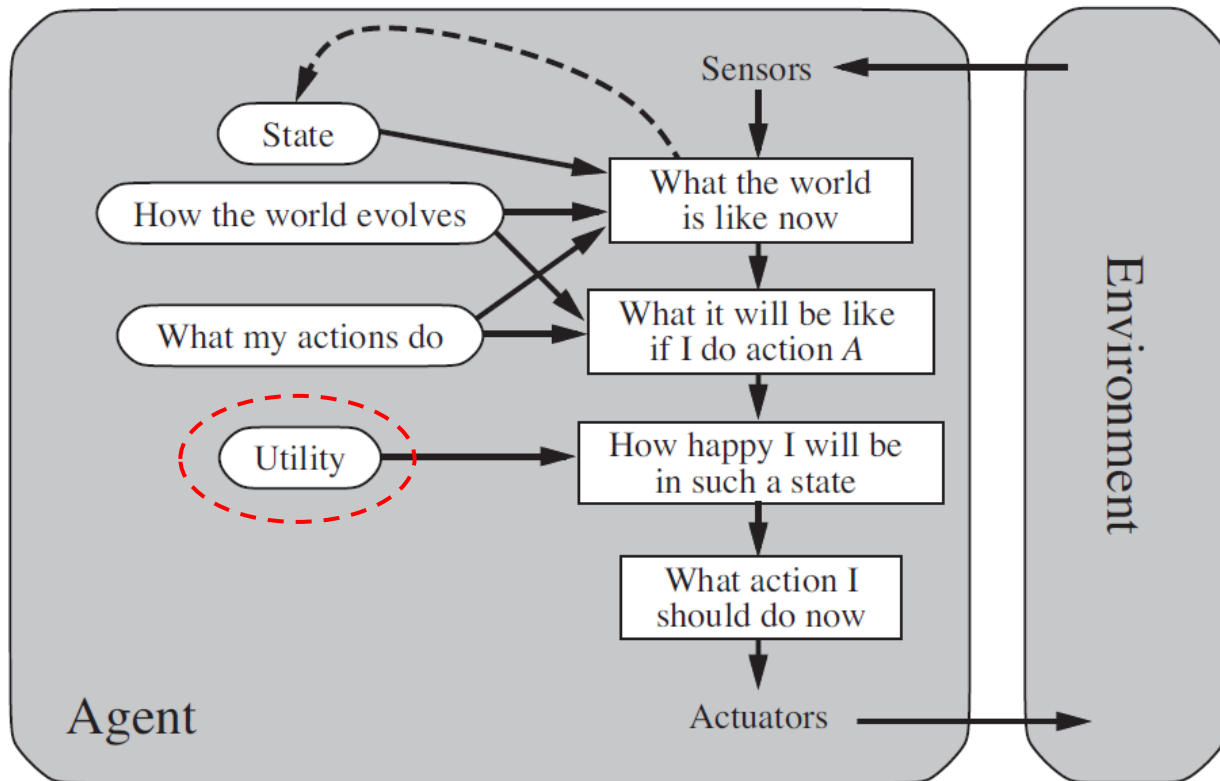
When the agent can do more than one action at the current state, it needs some sort of **goal** information that describes situations that are desirable.



Agent Program

4. Utility-based Agents

A more general performance measure should allow a comparison of different world states according to exactly how happy they would make the agent.



Properties of Environments

- **Fully observable** vs. **partially observable**

If an agent's sensors give it access to the complete state of the environment (that are relevant to the choice of action) at each point in time, then we say that the environment is **fully observable**.

An environment might be **partially observable** because of noisy and inaccurate sensors or because parts of the state are simply missing from the sensor data.

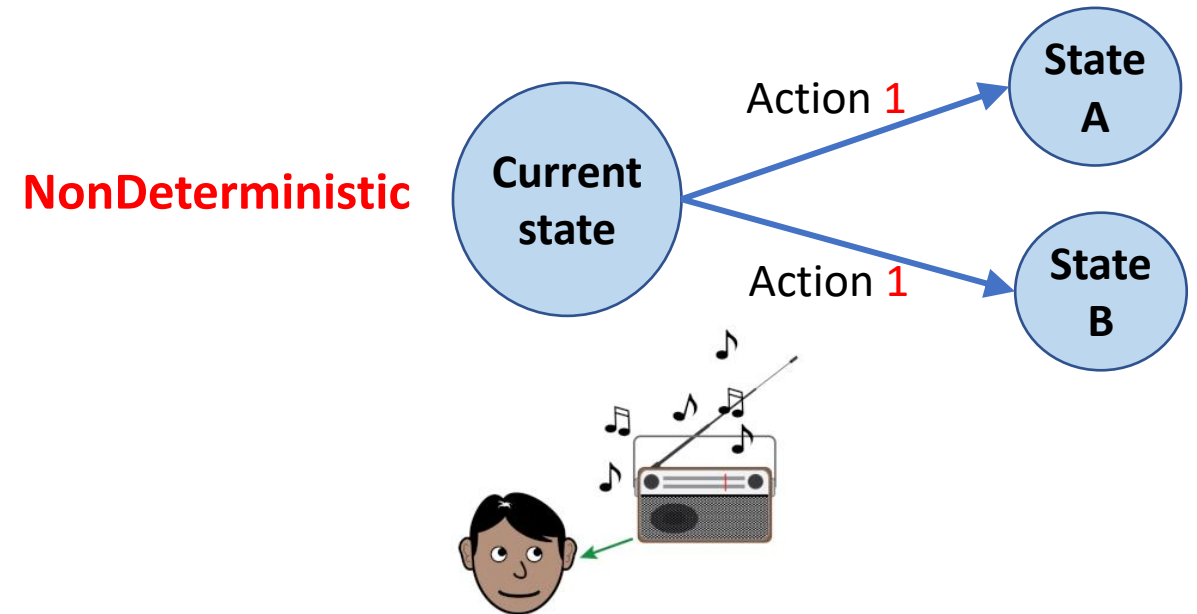
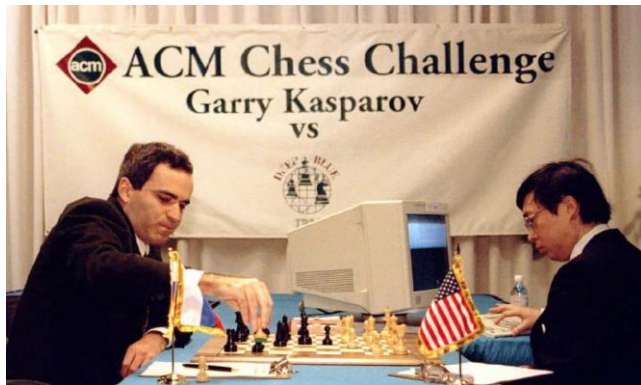
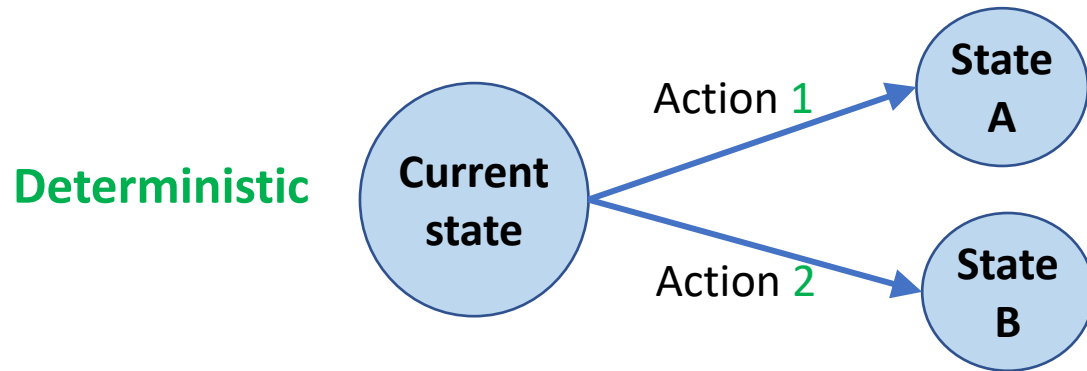
- **Single agent** vs. **multiagent**



Properties of Environments

- **Deterministic** vs. **stochastic**

If the next state of the environment is completely determined by the current state and the action executed by the agent, then we say the environment is **deterministic**; otherwise, it is **stochastic**.



The radio station: the listener is not aware of the next song.

Properties of Environments

- **Episodic** vs. **sequential**

Episodic: The agent's experience is divided into atomic episodes. In each episode the agent receives a percept and then performs a single action. Crucially, the next episode does not depend on the actions taken in previous episodes.

Episodic: Robot in assembly line



Sequential: Taxi driver



Properties of Environments

- **Static** vs. **dynamic**

If the environment can change while an agent is deliberating, then we say the environment is **dynamic** for that agent; otherwise, it is **static**.

Static : Backgammon



Dynamic : Part picking robot



Properties of Environments

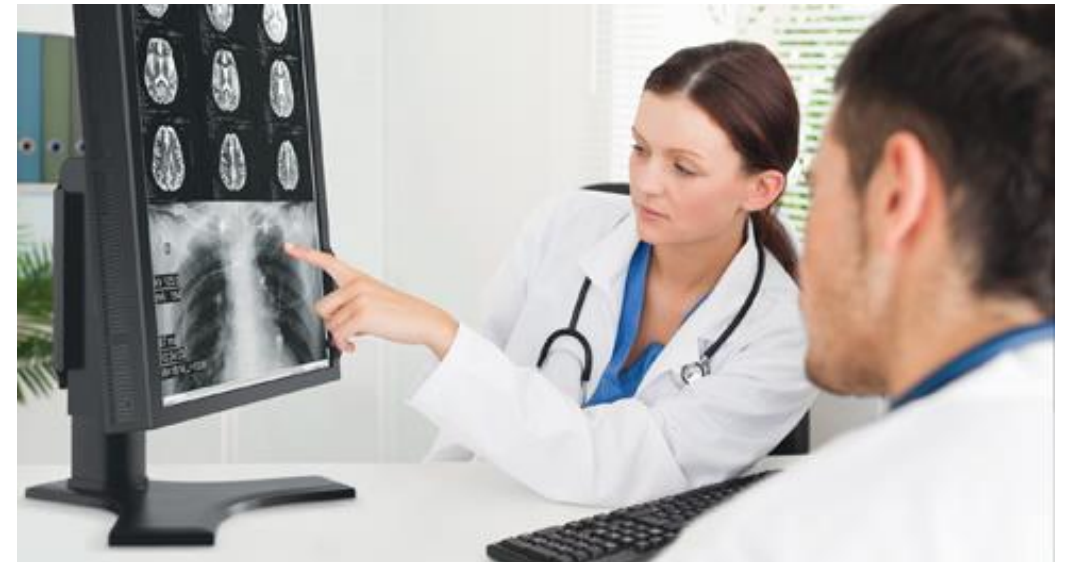
- **Discrete** vs. **continuous**

The environment is **discrete** if there are a limited number of distinct states, percepts, and actions. Otherwise, it is **continuous**.

Discrete



Continuous : Medical diagnosis



Properties of Environments

- **Known** vs. **unknown**

In a **known** environment, the outcomes for all actions are given. Obviously, if the environment is **unknown**, the agent will have to learn how it works in order to make good decisions.

Known : Checker



Unknown : Robot on unknown planet

